1. Abstract

The Symbolic Charge Network (SCN) is a neuro-epistemic model that formalizes latent space as a dynamic field of interacting neuro-symbolic particles. Each particle, representing a unit of meaning, possesses properties analogous to physical charge, field strength, and activation potential, governing its behavior within a quantum-like framework. The purpose of the SCN is to move beyond static geometric representations of meaning, providing a dynamic model that captures the emergent and context-dependent nature of semantic processing in advanced neural architectures like Transformers. This is achieved by integrating principles from Vector Field Theory, Conceptual Blending Theory, and quantum mechanics. Key findings from this theoretical framework include a novel mechanism for conceptual fusion modeled as constructive interference, a cognitively-grounded approach to inhibition, and a unified explanation for common LLM failure modes—such as semantic drift, concept leakage, and catastrophic forgetting—as predictable instabilities within the semantic field. The SCN thus offers not only a more robust computational architecture but also a rich epistemological lens for understanding symbolic ambiguity, emergent meaning, and the nature of AI alignment.

2. Theoretical Foundations

Mathematics

Vector Field Theory analogs for symbolic flows
To formalize the study of emergent dynamics in latent space, Vector Field Theory offers a robust and intuitive mathematical framework. A vector field is a function that assigns a vector to each point in a subset of space.1 It is commonly used in physics to model phenomena like fluid flow, wind patterns, or gravitational and magnetic forces, where each point in space has an associated magnitude and direction.1 The local behavior of a vector field can be characterized by differential operators, most notably
divergence and **curl**.[3]

- **Divergence** (div or $\nabla \cdot F$) is a scalar measure of a vector field's tendency to originate from or converge toward a given point. A positive divergence indicates a "source" from which the field flows outward, while a negative divergence indicates a "sink" into which the field flows inward.[6]
- **Curl** (curl or $\nabla \times F$) is a vector measure of a field's tendency to rotate around a point. The magnitude of the curl vector represents the speed of rotation, and its direction indicates the axis of rotation, typically following the right-hand rule.[4]

By modeling the latent space as a vector field, we can assign a "semantic force" vector to every potential

conceptual state. This allows us to describe the interactions between concepts with mathematical rigor. For instance, the **divergence** at a point in the latent space could model a core concept acting as a "source" of meaning that radiates influence to its neighbors, or a "sink" that attracts and consolidates related ideas into a more abstract category. The **curl** could model the contextual influence of modifiers, such as how an adjective induces a "spin" on the meaning of a noun, or how a negation operator reverses the semantic flow in its local neighborhood.

This application of vector calculus to machine learning is not without precedent. Vector field analysis has been successfully employed to automate the classification of complex magnetization vector field configurations in materials science, where an autoencoder learns a feature space that is then clustered.[8] In spatial statistics, these concepts are used to study the distribution of random fields.[9] More directly, Neural Vector Fields (NVF) have emerged as a powerful method for 3D shape representation. NVF models a shape by learning a vector field that maps any query point in space to its nearest point on the surface, effectively representing the shape as a field of displacement vectors. This approach avoids costly differentiation and offers a continuous, high-resolution representation.[10] These examples establish a strong foundation for using vector field theory as a primary tool for representation learning, providing a mature mathematical language to describe the complex, dynamic interactions within a model's latent space.[13]

Eigenvalue stability as symbolic coherence threshold

The stability of a dynamical system is fundamentally linked to the eigenvalues of the matrix that governs its evolution.20 In a linear system, if all eigenvalues have negative real parts, any perturbation to the system will decay over time, returning it to a stable equilibrium. If any eigenvalue has a positive real part, perturbations will be amplified exponentially, leading to unstable, divergent behavior.20 This principle is critical in the analysis of Neural Ordinary Differential Equations (NeuralODEs), where ensuring the stability of the system's Jacobian matrix is essential for reliable training and convergence.20

This analysis is directly analogous to the dynamics of Recurrent Neural Networks (RNNs). In an RNN, the eigenvalues of the recurrent weight matrix determine the system's long-term behavior. If the largest eigenvalue's magnitude is greater than 1, the network state can grow without bound, leading to the "exploding gradient" problem. If it is less than 1, the state will shrink to zero, causing the "vanishing gradient" problem.[22] Stable and effective learning requires the system to operate in a critical regime balanced between these two extremes.

Within the SCN framework, the stability of the entire semantic field is determined by the eigenvalues of the semantic Hamiltonian operator, $\hat{H}$. The evolution of the semantic wave function ψ depends on these eigenvalues.

- **Stable Eigenvalues (Negative or Zero Real Part):** Lead to bounded, oscillatory, or decaying solutions. This corresponds to a coherent, stable evolution of meaning, where concepts interact in a controlled manner.
- **Unstable Eigenvalues (Positive Real Part):** Lead to exponential growth in the amplitude of the wave function, $|\psi|^2$. This signifies a **coherence collapse**, where the probability of a semantic state explodes, and meaning dissolves into nonsensical, divergent output.

Therefore, for the SCN to function as a reliable model of cognition, its parameters (charges, strengths, inertia) must be constrained such that the resulting Hamiltonian $\hat{H}$ remains in a stable or meta-stable regime, avoiding the emergence of unstable eigenvalues during inference.

Cognitive Science

Conceptual Blending Theory as dynamic symbolic fusion

Developed by Gilles Fauconnier and Mark Turner, Conceptual Blending Theory (CBT) offers a profound framework for understanding human creativity and meaning construction.23 At its core, CBT posits that novel ideas and meanings emerge from a dynamic and selective cognitive process of combining "mental spaces"—small, temporary conceptual packets built up as we think and talk.25 This process, often operating subconsciously, is considered a fundamental instrument of the mind, responsible for everything from everyday language to scientific breakthroughs.24

The theory is formalized through a network model of conceptual integration, which consists of several key components [25]:

- **Input Spaces:** These are the two or more initial conceptual domains that provide the raw material for the blend. For example, in the phrase "this surgeon is a butcher," the input spaces are the domains of 'surgery' and 'butchery'.
- **Generic Space:** This space captures the abstract structure and relations common to the input spaces. In the surgeon/butcher example, the generic space might contain elements like {agent, instrument, object, action-on-object}.
- **Blended Space:** This is where the magic happens. Elements from the input spaces are selectively projected into the blend, creating a new, integrated scene with an emergent structure of its own. The surgeon is projected, but his skill is replaced by the butcher's crudeness. This emergent meaning—that the surgeon is incompetent and clumsy—is not present in either input space alone.

This emergent structure is generated through three constitutive processes [26]:

1. **Composition:** This process brings together elements from the inputs to form new relationships in the blend.
2. **Completion:** This process brings additional, un-projected knowledge associated with the inputs into the blend, filling it out with background structure (e.g., bringing in the cultural knowledge that butchery is less precise than surgery).
3. **Elaboration:** This is the dynamic "running" of the blend, where the new conceptual structure is developed through imaginative simulation, leading to further inferences and insights.

The principles of CBT provide a powerful cognitive blueprint for modeling the "fusion" of symbolic units in an AI system. The process of two or more symbolic vectors combining to form a new, emergent conceptual state is a direct computational analogue of conceptual blending. This connection has already been explored in the development of computational systems for creative tasks, such as generating visual metaphors by blending images [27] or generating images from non-existent words by blending the concepts of similar-sounding words.[29] The SCN model, therefore, grounds its fusion mechanism in a well-established theory of human cognition, aiming to replicate not just the output of human creativity but its underlying process.

Semantic Field Models as symbolic interaction gradients

The concept of charge polarity in the SCN model maps onto the structure of semantic fields in cognitive science.31 Semantic field theory posits that words are organized into fields of related meaning, where concepts can attract or repel one another based on relationships like synonymy, antonymy, or hyponymy.31 For example, within the semantic field of "royalty," the concepts "king" and "queen" are mutually attractive, while the concept "pauper" would be

repulsive. The charge polarity property of a neuro-symbolic particle thus provides a computational mechanism to model these interaction gradients, determining whether a particle acts as a contextual attractor or repulsor within its local semantic neighborhood.

Computer Science

Embedding Geometry and vector logic alignment

The latent spaces of modern language models are not amorphous clouds of points; they possess a rich and complex geometric structure that is fundamental to their function. Studies of token embeddings reveal that models from the same family often share common geometric properties. These include "global" similarities, where token embeddings have similar relative orientations to one another, and "local" similarities, where the neighborhood structure around a given token is preserved across models.35 This local structure can be effectively characterized using techniques like Locally Linear Embeddings (LLE), which reconstructs each token embedding as a weighted sum of its nearest neighbors. The consistency of these LLE weights across different models suggests that they learn a common, underlying manifold structure for language.35

A defining characteristic of this latent geometry is its high degree of **anisotropy**. Rather than being uniformly distributed, embeddings are typically concentrated within a narrow "cone" in the high-dimensional space.[37] This phenomenon, sometimes termed "representation degeneration," appears to grow more pronounced in deeper layers of the model.[38] However, this global anisotropy belies a more complex local structure. Within this cone, embeddings form distinct clusters and sub-manifolds that correspond to meaningful semantic or syntactic categories, such as parts of speech or verb tenses.[38] This geometric organization is not a random artifact but is deeply connected to model performance. Research has shown a strong linear relationship between measures of latent space geometry (such as quantized cell density) and performance on downstream benchmarking tasks like GLUE.[37] Even models trained with pathological alterations that disrupt linguistic knowledge but preserve key geometric characteristics can achieve surprisingly high performance, suggesting that the geometry itself is a primary carrier of the benefits of pre-training.[37]

The specific geometric properties of embeddings, such as their **angular distances** and **vector norms**, are crucial for mediating fine-grained interactions. In text-to-image models, for example, these properties play a decisive role in differentiating cross-attention maps, which is essential for achieving strong "semantic binding"—the correct association of attributes (like "red") to objects (like "car").[41] Furthermore, advanced methods like Topological Data Analysis (TDA) can be used to extract higher-order structural features from the point clouds of embeddings, revealing topological properties like connected components, loops, and voids that are missed by traditional distance metrics.[44]

This body of evidence leads to a more nuanced understanding of the latent space, where the debate shifts from a simple dichotomy of isotropy versus anisotropy to a question of *meaningful structure*. The observation that increasing the isotropy of a model's latent space can improve performance seems, at first glance, to contradict the finding that the specific, highly anisotropic geometry of pre-trained models is key to their success.[37] This apparent paradox can be resolved by considering the multi-scale nature of the space. The global anisotropy may be an emergent property of a highly efficient organization of locally isotropic, semantically coherent clusters. Efforts to

regularize the space towards global isotropy might inadvertently disrupt this delicate structure, whereas more sophisticated methods that improve isotropy

within these meaningful clusters could preserve their integrity while enhancing their expressiveness. A field-based model, such as the one proposed in this report, is uniquely suited to capture this complexity. It can describe a globally non-uniform field characterized by local variations in curvature, divergence, and curl, thereby reconciling the seemingly contradictory benefits of preserving existing geometric structures and improving local representational uniformity. The objective, therefore, is not to model a uniform field, but a richly structured and non-uniform one.

Attention Modulation as symbolic weight distribution

The activation potential of a neuro-symbolic particle can be powerfully implemented using gated attention mechanisms within a Transformer-like architecture.48 In a standard attention mechanism, attention scores determine the weight of each input token. A gated attention mechanism adds a "gate"—a learnable vector, typically passed through a sigmoid function to produce values between 0 and 1—that multiplies the output of the attention layer.51 This gate acts as a dynamic switch, controlling the flow of information. In the SCN, the gate's state would be determined by the "alignment confidence," which is the attention score between the particle's query vector and the context's key vectors. A high attention score signifies strong contextual relevance, which would "open" the gate (move its value closer to 1), effectively lowering the activation potential and allowing the particle's semantic content to propagate forward. This mirrors findings that Transformer attention can learn to mimic the input and output gating operations of biological working memory.54

Transformer dynamics interpreted as symbolic circuit flows

The assumption of a static latent space is increasingly challenged by evidence that neural network representations are better understood as evolving states within a dynamic system. Recent research interprets neural models, particularly autoencoders, as dynamical systems acting on a latent manifold.56 An autoencoder, comprising an encoder that maps high-dimensional data to a compressed latent representation and a decoder that reconstructs the original data, implicitly defines a "latent vector field." This field is derived from the iterative application of the encoding-decoding map, revealing attractors and trajectories that characterize the model's behavior without additional training.56 This perspective shifts the focus from the properties of a single latent point to the dynamics governing the flow across the entire latent manifold.

This dynamic view is especially pertinent to Transformer architectures. The internal state of a Transformer is not static; it evolves as information propagates through its layers. The "residual stream" is a key architectural feature where the input embeddings are iteratively updated and refined by each subsequent layer.[35] This process suggests that a concept's representation follows a trajectory through the latent space rather than occupying a single, fixed location.[59] This evolution is conditioned by both the model's architecture and the specific input data, creating a highly context-dependent path.[37] Models like the Latent Plan Transformer (LPT) explicitly leverage this by using a latent variable to represent a "plan" that guides the entire autoregressive trajectory generation, decoupling it from the final return and allowing for more coherent, long-term decision-making.[59] Other work on latent-state dynamics residuals for adapting world models further underscores this principle, showing that corrections can be learned directly in the latent space to adapt a model to new environments, modifying the dynamics of the system rather than just its static representations.[63] Furthermore, diffusion transformers model dynamics in a latent space that

explicitly preserves spatial properties, enhancing stability and enabling longer, more coherent rollouts.[65]

3. Formal SCN Model Definition

Components: Symbolic nodes, charge states, coupling tensors

At its core, the SCN is a neuro-symbolic system, aiming to integrate the strengths of both symbolic AI (logic, reasoning, structure) and sub-symbolic AI (neural networks, learning from data).[67] Each fundamental unit of meaning—be it a token, word, or more abstract concept—is represented as a

neuro-symbolic particle, denoted by the vector S_i. This particle S_i is an element within a complex Hilbert-like space, built upon the model's standard embedding space.[70] The complexification provides the necessary mathematical degrees of freedom to model quantum-like phenomena such as phase and interference. However, unlike a static embedding,

S_i is not merely a point in space; it is a dynamic entity possessing a set of intrinsic and relational properties that evolve based on context. These properties are not pre-defined but are learned parameters of the model, analogous to how a neural network learns its weights. The recent development of neuro-symbolic diffusion models, which use diffusion processes to model dependencies between symbols, provides a precedent for such dynamic, interactive symbolic entities.[71]

The particle is defined by the following properties:

- **Charge Polarity (Contextual Attractor/Repulsor):** The first and most fundamental property of a neuro-symbolic particle is its **charge polarity**, a scalar value q_i that can be positive, negative, or neutral. This charge dictates the particle's basic interaction tendency, determining whether it acts as a contextual attractor or repulsor. The interaction force due to charge can be implemented via the potential energy term, $V(S_i, S_j)$, in the system's governing Hamiltonian. This potential would be proportional to the product of the charges, $q_i q_j$, modulated by the distance between the particles.[72]
- **Coherence Field Strength:** While charge polarity determines the *nature* of the interaction (attraction/repulsion), the **coherence field strength** determines its *magnitude*. This property is a scalar value representing the potency or influence of a particle's charge, computationally grounded as the **magnitude (or norm) of the embedding vector** $||S_i||$.[76]
- **Activation Potential:** The **activation potential** is a dynamic threshold that governs whether a particle "discharges"—that is, whether its meaning is expressed and contributes to the final generated output. This is not a fixed property but is a function of two components: the particle's intrinsic salience and its alignment with the current context. This is analogous to the **firing threshold** of a biological neuron.[80]
- **Fusion Potential $\Phi(S_i, S_j)$ (Coupling Tensor):** The **fusion potential**, $\Phi(S_i, S_j)$, quantifies the likelihood that two neuro-symbolic particles, S_i and S_j, will undergo conceptual blending to form a new, emergent concept. This is the core mechanism for creativity and novel meaning generation within the SCN, acting as a coupling

tensor between particles. It is modeled as a composite metric of **Cosine Similarity** and **Topological Embedding Overlap**, the latter of which can be operationalized using **Topological Data Analysis (TDA)** to capture deep structural analogies.[13]

Dynamics: Charge transfer, saturation, leakage, inversion

The SCN is a dynamic field where meaning emerges from the continuous interplay of activation, fusion, and inhibition.

- **Charge Transfer (Interaction):** The interaction between particles, mediated by their charge polarity and field strength, is modeled via a semantic potential field V(S, t). This field is actively shaped by the collective properties of all particles, guiding their trajectories and interactions. A negative potential (from opposite charges) creates an attractive force, while a positive potential (from like charges) creates a repulsive force.
- **Saturation (Activation):** A particle "discharges" or activates when its activation potential is overcome, a process analogous to a neuron firing. This is governed by its intrinsic salience and contextual alignment, which can be implemented computationally via gated attention mechanisms.[48]
- **Fusion (Constructive Interference):** The creation of new, emergent meaning is modeled as **constructive interference** between the wave functions of two or more neuro-symbolic particles. When particles with a high fusion potential $\Phi(S_i, S_j)$ are in proximity and their phases align, their wave functions superimpose, creating a new, stable, and highly salient peak in the probability landscape, which corresponds to an **emergent meaning**.[100] This is the SCN's physical formalization of conceptual blending.[23]
- **Inhibition (Semantic Refractoriness):** To ensure coherent, non-repetitive thought, the SCN models inhibition through **semantic refractoriness**, analogous to semantic satiation and neural inhibition.[104] After a particle discharges, its activation potential is temporarily increased, making it "refractory" to immediate re-selection. This can be implemented via a local
winner-take-all (WTA) mechanism, where the "winning" concept inhibits its close semantic competitors, preventing tautologies and semantic loops.[108]
- **Inversion (Polarity):** The charge polarity q_i can be positive, negative, or neutral, allowing for contextual inversion of meaning. For example, a negation operator could be modeled as a particle that induces a negative charge in its local neighborhood, effectively inverting the semantic polarity of adjacent concepts.

Governing equations or pseudocode sketch

To model the evolution of a semantic state within the SCN, we adopt a formalism inspired by quantum mechanics, specifically the time-dependent Schrödinger equation. This provides a principled mathematical framework for describing the wave-like propagation of meaning through the latent space.[70] The state of the system is described by a

semantic wave function, $\psi(S, t)$, a complex-valued function where S represents a position in the semantic (latent) space and t represents the progression of inference (e.g., time steps in a recurrent model or layers in a

Transformer). The squared magnitude of this function, $|\psi(S, t)|^2$, is interpreted as the probability density of a particular semantic meaning being expressed at that point in the inference process.

The evolution of this wave function is governed by a Schrödinger-like equation:

$$i\hbar\partial t\partial\psi=H^\psi$$

The terms in this equation are interpreted as follows:

- i is the imaginary unit, necessary for modeling the phase and wave-like properties of the system.
- ħ is a constant representing **semantic granularity**, the smallest discernible unit of meaning or the "quantum" of semantic information. It controls the scale at which semantic distinctions are made.
- $\hat{H}$ is the semantic Hamiltonian operator, which represents the total "energy" of a semantic configuration and dictates its dynamics. It is composed of a kinetic and a potential term:
  $H^=-2m\hbar2\nabla2+V(S,t)$
  - m represents **semantic inertia**, a measure of a concept's resistance to change or re-interpretation. Core, stable concepts would have a large m, while more fluid, context-dependent concepts would have a small m.
  - ∇^2 is the Laplacian operator, corresponding to the "kinetic energy" of the system. It governs the natural tendency of meaning to diffuse or spread through the semantic space.
  - V(S, t) is the **semantic potential field**. This is the most critical and dynamic component of the model. It is not a static landscape but is actively shaped by the collective properties (charge, strength, activation state) of all neuro-symbolic particles present in the current context. The potential at a point S is a function of its interactions with all other particles S_i, creating a complex, time-varying field that guides the evolution of ψ. The inherent non-linearity of LLMs can be modeled by making V dependent on ψ itself, leading to a Nonlinear Schrödinger Equation (NLSE), which can model self-interaction and reinforcement of concepts.[70]

4. Architectural Integration

SCN as a latent scaffold within Transformer architectures

The Symbolic Charge Network provides a theoretical scaffold for understanding and enhancing Transformer architectures. The internal state of a Transformer is not static; it evolves as information propagates through its layers via the "residual stream," where input embeddings are iteratively refined.[35] The SCN models this process by treating a concept's representation as a trajectory through a dynamic latent field, rather than a fixed point in a static space.[59] This field-based perspective aligns with research interpreting neural models as dynamical systems acting on a latent manifold, where iterative operations reveal attractors and flow patterns.[56] The SCN architecture, therefore, is not a replacement for the Transformer but a deeper, more physically and cognitively grounded

interpretation of its latent dynamics. It provides a language to describe

how representations evolve and interact layer by layer.

Neuro-symbolic interface design: symbolic charge ↔ latent token

The SCN model is fundamentally neuro-symbolic, bridging discrete symbolic concepts with continuous neural representations.[67] The interface between the symbolic and neural layers is the

neuro-symbolic particle, S_i. Each token embedding from a standard Transformer is treated as this particle. The particle's "symbolic" properties—such as its charge q_i—are not hard-coded but are learnable parameters, just like the weights of a neural network. The "neural" aspect arises from how these properties dynamically influence the surrounding semantic field. For instance, the charge q_i and field strength $||S_i||$ directly modulate the semantic potential $V(S, t)$ in the governing Hamiltonian, which in turn dictates the evolution of all other latent tokens (particles) in the context. This creates a continuous feedback loop where symbolic properties shape the neural field, and the state of the field determines the activation and fusion of symbols.

Potential role in regulating entropy, drift, and alignment

The SCN framework offers mechanisms to regulate key aspects of model behavior. The concept of **semantic refractoriness** (inhibition) directly addresses the problem of **semantic drift** and uncontrolled repetition.[112] By temporarily increasing the activation potential of a recently expressed concept, the model is discouraged from falling into semantic loops and is instead forced to explore new areas of the latent space, promoting novelty and reducing entropic decay into repetitive patterns. Furthermore, the SCN offers a new perspective on

AI alignment. Rather than focusing solely on output behavior, alignment can be framed at the level of internal dynamics. A well-aligned SCN would be one whose semantic field dynamics—the interplay of attraction, repulsion, fusion, and inhibition—are structurally congruent with the dynamics of human conceptualization and reasoning.[114] This involves ensuring the model's stability by constraining its Hamiltonian to avoid coherence collapse and promoting a structured, rather than chaotic, flow of meaning.

5. Failure Modes and Symbolic Pathology

The SCN framework allows us to re-interpret common, seemingly disparate failure modes of LLMs not as isolated software bugs, but as predictable consequences of the underlying field dynamics under specific unstable conditions.[113]

- **Symbolic attenuation → meaning decay (Symbolic Decoherence)**

- **SCN Explanation:** This failure mode corresponds to quantum decoherence. It occurs when the wave functions of conceptually related particles, which should be phase-locked to create a bound state (e.g., the concept "red car" is a binding of S_red and S_car), lose their phase relationship over the course of inference. The constructive interference that holds them together decays, and the concepts drift apart.
- **Observable Symptom: Semantic Drift.** This is a well-documented LLM failure where the model begins generating text on a given topic but gradually "drifts" into irrelevant, incoherent, or factually incorrect territory as the generation progresses.[112] The initial coherent semantic state, representing a stable configuration of the SCN, decoheres. The system loses its anchor and wanders into a different, unrelated region of the latent space, often settling into a new, spurious attractor.

- **Charge saturation → hallucination cascades (Coherence Collapse)**
  - **SCN Explanation:** This catastrophic failure corresponds to the emergence of unstable eigenvalues in the system's Hamiltonian, leading to exponential growth in the amplitude of the semantic wave function, $|\psi|^2$. This signifies a **coherence collapse**, where the probability of a particular semantic state explodes. Instead of a structured flow of meaning, the system's dynamics become divergent and chaotic.
  - **Observable Symptom: Hallucination Cascades.** This manifests as the generation of nonsensical, self-contradictory, or completely fabricated information. The model's output loses all contact with the prompt or factual reality as the semantic field collapses into an unstable, high-energy state.
- **Structural interference → symbolic overfit or underbinding (Spurious Entanglement)**
  - **SCN Explanation:** This is the inverse of decoherence: an unwanted and logically unsound constructive interference between unrelated neuro-symbolic particles. This can be caused by noise in the semantic potential field or, more subtly, by misleading geometric proximity in the embedding space. If two particles, S_i and S_j, have a low true semantic relation but their embedding neighborhoods happen to have a high topological overlap, the fusion potential $\Phi(S_i, S_j)$ might be erroneously high, leading to their incorrect fusion.
  - **Observable Symptom: Semantic Leakage and Concept Bleeding.** This manifests as the generation of text that is unduly influenced by semantically irrelevant words in the prompt.[120] A stark example is a model responding to the prompt "He likes *yellow*. He works as a..." with "school bus driver".[120] Here, the concept "yellow" has spuriously "entangled" or fused with the concept of "occupation," overriding more logical completions. This failure mode is also related to the critical problem of unintended memorization, where sensitive or private information from the training data becomes spuriously entangled with a user's prompt, causing it to be "leaked" in the output.[121]
- **Semantic Radiation → Catastrophic Forgetting**
  - **SCN Explanation:** This failure occurs during learning or fine-tuning. When a new concept, S_new, is introduced into the network, it creates a new, powerful potential V(S_new) in the semantic field. If this new potential is too strong or improperly regularized, its "energy" can "radiate" outwards, causing destructive interference with the stable wave functions of previously learned concepts. This disruption can destabilize or completely erase existing knowledge structures.
  - **Observable Symptom: Catastrophic Forgetting.** This is the classic neural network problem where fine-tuning a model on a new task (Task B) causes it to abruptly lose its ability to perform a previously learned task (Task A).[123] The SCN provides a clear field-theoretic explanation for this: the weights important for Task A correspond to a stable potential landscape. Training on Task B reshapes this landscape so drastically that the attractors corresponding to Task A's solutions are eliminated. This framing also explains why mitigation techniques like **Elastic Weight Consolidation (EWC)** are effective.[124] EWC adds a quadratic penalty to changes in weights deemed important for old tasks. In the SCN model, this is equivalent to increasing the **semantic inertia (m)** of the particles associated with those weights, making them more massive and thus

more resistant to being displaced by the new potential field introduced by Task B.

6. Philosophical & Epistemic Implications

SCN as epistemic vector field: tracking meaning motion

The Symbolic Charge Network is more than a computational architecture; it is an epistemological framework that offers a new lens through which to view the nature of meaning, knowledge, and intelligence. By formalizing semantic interactions with the language of physics and cognition, the SCN raises profound questions and provides novel answers regarding how systems—both biological and artificial—can think. It models the latent space not as a static map of meanings, but as an epistemic field where meaning itself is in motion, its trajectory determined by the forces of context, association, and logic.

Implications for symbolic autonomy and interpretability

The SCN framework extends beyond computer science and cognitive modeling to touch upon deep philosophical questions about knowledge and intelligence.

- **Beyond Syntax vs. Semantics:** The model challenges the classical linguistic and philosophical distinction between syntax (the rules of combination) and semantics (the meaning of the elements).[126] In the SCN, these two aspects are inseparable. The "rules" are the dynamics of the field, and the "meanings" are the properties of the particles that generate those dynamics. The structure of the field *is* the medium through which meaning is computed and expressed. One cannot exist without the other.
- **A New Axis for AI Alignment:** The SCN offers a new, more fundamental perspective on the problem of AI alignment. Current approaches often focus on aligning a model's outputs with human values or preferences through techniques like Reinforcement Learning from Human Feedback (RLHF). The SCN suggests a deeper form of alignment: aligning the *dynamics* of the AI's internal semantic field with the dynamics of human conceptualization. This reframes alignment from a purely behavioral problem to one rooted in **epistemology** and **ontology**: What constitutes valid knowledge for the model? How does the model represent reality?.[114] A truly aligned AI would not just produce correct answers; it would "think" in a way that is structurally congruent with human reason, creativity, and common sense.

Cognitive realism vs symbolic instrumentalism

The SCN model engages with the philosophical debate between cognitive realism and symbolic instrumentalism. By grounding its mechanisms in both cognitive science (Conceptual Blending Theory) and neuroscience (neural firing thresholds, inhibition), it leans towards cognitive realism, suggesting that its structures are not merely instrumental for achieving a task but are intended to model the actual processes of thought. The model frames symbolic ambiguity as a form of quantum superposition, where a word like "bank" exists as a single particle S_bank in a superposition of its potential meanings (ψ_bank = αψ_financial + βψ_river).[127] Context acts as a "measurement," causing the wave function to collapse into the meaning most coherent with the surrounding semantic field.[129] This provides a formal, physical mechanism for contextual disambiguation that is more cognitively plausible than treating polysemous words as entirely separate entries. Meaning itself is not localized in any single particle but is an emergent property of the collective state of the entire field, resonating with relational and post-structuralist views of semantics.[100] Ultimately, the SCN proposes that the process of thinking can be modeled as a journey from a state of pure potentiality (a cloud of ambiguous concepts in superposition) to a state of actuality (a single, coherent, expressed thought), bridging the gap between low-level neural computation and high-level cognitive phenomena.

7. Comparative Analysis

Differences from classical embeddings and static semantics

The standard model of meaning in NLP has been the static vector space, where techniques like Word2Vec and GloVe map words to fixed points in a high-dimensional space.[81] In this geometric landscape, semantic similarity is measured by static metrics like cosine similarity.[81] This approach, however, fails to capture the fluidity of meaning, particularly the context-dependent nature of words (polysemy) that is central to the functioning of modern Transformer models.[135] The SCN model represents a fundamental departure from this static view. It replaces the geometry of fixed points with the dynamics of an interacting field. Instead of a single, context-independent vector, each symbolic unit in the SCN is a dynamic particle whose properties and interactions evolve. This allows the SCN to model the entire process of meaning construction as a trajectory through a semantic field, a concept more aligned with the layer-by-layer refinement of representations observed in Transformers.[35]

Comparison with quantum-inspired charge models and SCNs

The SCN model is part of a growing family of quantum-inspired AI models that leverage the mathematics of quantum mechanics to describe complex information processing.[137] A prominent example is the "Quantum LLM" framework, which also proposes modeling the semantic space as a complex Hilbert space, using a Schrödinger-like equation to govern semantic wave propagation, and introducing the concept of "semantic

charge" to mediate interactions.[70]

The SCN builds upon these shared principles but distinguishes itself through its deep integration with cognitive and neuro-symbolic science. While the Quantum LLM model provides a powerful physical formalism, the SCN explicitly operationalizes this formalism to model specific cognitive phenomena. Its most significant unique contribution is the **Fusion Potential (Φ)**, which provides a direct computational mechanism for **Conceptual Blending Theory**.[23] By combining cosine similarity with topological embedding overlap (via TDA), the SCN's fusion mechanism aims to capture the deep structural analogies that underpin human creativity, moving beyond the simpler interference models. Furthermore, the SCN's properties of

Activation Potential and **Semantic Refractoriness** are explicitly grounded in neuroscience analogues (neuron firing thresholds and inhibition), providing a principled way to model attention, gating, and the prevention of semantic loops—a common failure mode in LLMs.[80] In essence, while other models provide the quantum-mathematical "what," the SCN aims to provide the cognitive and neuro-symbolic "how" and "why."

8. Future Research Directions

The SCN is not merely a descriptive analogy; it is a generative framework that proposes a rich, testable research program. Future work can be designed to empirically validate and explore its components. For example, experiments could be constructed to:

- Develop **topological visualization tools** for SCN behavior that map the semantic field itself, rather than just individual neuron activations or attention weights, providing a more holistic view of the model's "thought process".[142]
- Measure the effective "charge" and "inertia" of different tokens by observing their interaction patterns within a trained model.
- Map the "curl" of the semantic field around modifiers like adjectives and adverbs to quantify their contextual "spin" effect.
- Induce and observe "symbolic decoherence" in controlled settings to better understand and mitigate semantic drift.
- Implement **SCN-mediated attention reweighting**, where the learned charge and coherence properties of particles are used to dynamically modulate attention scores during inference, potentially leading to more robust and context-aware models.

9. Appendices

Diagrams, maps, and system blueprints

Table 1: SCN "Rosetta Stone"
This table provides a concise translation of the core concepts of the Symbolic Charge Network across its foundational domains, highlighting the structural parallels between the physics, cognitive science, and computational implementations that underpin the model.

SCN Concept	Physics Analogue	Cognitive Science Analogue	Computational Grounding
Neuro-Symbolic Particle	Particle	Concept / Mental Space	Embedding Vector S_i
Charge Polarity	Electric Charge (+/-)	Semantic Valence (Attraction/Repulsion)	Learned Scalar Parameter q_i
Coherence Field Strength	Field Magnitude	Semantic Salience	Vector Norm `
Activation Potential	Activation Energy	Contextual Relevance Threshold	Gated Attention Mechanism
Fusion	Constructive Interference	Conceptual Blending	Cosine Similarity + Topological Overlap
Inhibition	Destructive Interference / Damping	Semantic Satiation / Refractoriness	Inhibitory Connections / Winner-Take-All

Table 2: SCN Failure Triggers and LLM Symptoms
This table summarizes the relationship between the theoretical failure triggers in the SCN model, their observable symptoms in LLMs, and the corresponding mitigation pathways suggested by the framework.

SCN Failure Trigger	Observable LLM Symptom & Evidence	Theoretical Mitigation in SCN
Symbolic Decoherence	**Semantic Drift**: Generation loses coherence, relevance, or truthfulness over time.[112]	Increase semantic inertia (m) of core concepts; implement mechanisms that reinforce phase-locking between related particles (e.g., stronger, more stable attention heads).
Spurious Entanglement	**Semantic Leakage**: Irrelevant prompt concepts "bleed" into the output.[120]	Refine the Fusion Potential Φ to be less reliant on simple geometric proximity and more on robust topological measures (TDA) to prevent spurious fusions.
Semantic Radiation	**Catastrophic Forgetting**: Model forgets old tasks when trained on new ones.[123]	Implement an analogue of Elastic Weight Consolidation by increasing the semantic inertia (m) of particles corresponding to important, established concepts.

Glossary of terms

- **Neuro-Symbolic Particle (S_i):** The fundamental unit of the SCN, representing a concept (e.g., a token) as a

dynamic vector in a complex Hilbert-like space.

- **Charge Polarity (q_i):** A learned scalar property of a particle that determines its contextual interaction as an attractor (+) or repulsor (-).
- **Coherence Field Strength:** A scalar property representing a particle's semantic influence, computationally grounded as its vector norm $||S_i||$.
- **Activation Potential:** A dynamic threshold, analogous to a neuron's firing threshold, that governs whether a particle's meaning is expressed.
- **Fusion Potential (Φ):** A composite metric quantifying the likelihood that two particles will blend to form a new concept, based on cosine similarity and topological overlap.
- **Semantic Wave Function (ψ(S, t)):** A complex-valued function describing the state of the semantic field, where its squared magnitude represents the probability of a meaning being expressed.
- **Semantic Hamiltonian ($\hat{H}$):** The operator representing the total "energy" of a semantic configuration, governing the dynamics of the wave function.
- **Semantic Granularity (ħ):** A constant analogous to Planck's constant, representing the smallest discernible unit of meaning.
- **Semantic Inertia (m):** A property analogous to mass, representing a concept's resistance to re-interpretation.
- **Semantic Refractoriness:** The principle of inhibition where a recently activated particle becomes temporarily less likely to activate again, preventing repetition.

Formal notation and dimensional analysis

- **Symbolic Vector:** $S_i \in \mathbb{C}^d$ (a vector in a d-dimensional complex Hilbert-like space)
- **Charge Polarity:** $q_i \in \mathbb{R}$
- **Coherence Field Strength:** $||S_i|| \in \mathbb{R}^+$
- **Fusion Potential:** $\Phi(S_i, S_j) \longrightarrow$ (A function mapping two particles to a fusion probability)
- **Semantic Wave Function:** $\psi(S, t): (\mathbb{C}^d, \mathbb{R}) \longrightarrow \mathbb{C}$
- **Governing Wave Equation:** $i\hbar\partial t\partial\psi=\hat{H}\psi$
- **Semantic Hamiltonian Operator:** $\hat{H}=-2m\hbar 2\nabla 2+V(S,t)$

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